Aden James Calinisan

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Embedded Software Engineering | C/C++ • Python • Zephyr RTOS • Linux

EDUCATION

University of California, Irvine (GPA 3.846) **December 2026**
Bachelor of Science, Computer Engineering
Florida State College at Jacksonville (GPA 4.000) **May 2023**

SKILLS

- **Programming:** C, C++, Python (TensorFlow, PyTorch), Java
- **Embedded:** Zephyr RTOS, Oscilloscope, SPI, I²C, UART, GPIO, DeviceTree, MCU Development (ie ARM-CMSIS)
- **Tools:** Linux, CMake, WSL, Git, GitLab, VS Code, TensorFlow Lite, NN Model Compilation
- **Miscellaneous:** NAR HP Rocketry Level 1, Microsoft Excel, Solidworks, MATLAB, OrcaSlicer, ArduPilot

WORK EXPERIENCE

Applications Engineering Intern | *Syntiant - Irvine, CA* **February 2026 – Present**

- Port proprietary SDK firmware to Zephyr RTOS in C, adapting SPI/I²C communication interfaces and configuring hardware peripherals through DeviceTree overlays
- Benchmark MCU/NPU inference performance across multiple architectures, analyzing energy efficiency (µJ/inference), throughput (inferences/sec), and memory utilization (ie. ARM-Cortex M, Nordic nRF, NXP MCX)
- Build, flash, and debug embedded firmware across various vendor SDKs and development environments
- Convert and validate neural network models (CNN, GRU, BiGRU, MHA) for embedded NPU inference using Linux-based model conversion tools through WSL
- Debug I²C communication and validate clock frequencies using an oscilloscope and multimeter

Barista & Finance | *Boba Bee - Lake Forest, CA* **January 2024 – February 2026**

- Communicate with event coordinators and delegate tasks to team members while leading catering events
- Create quarterly financial reports for 4 Boba Bee locations cutting accounting fees by 80%

Finance Assistant | *KayCal Ventures LLC - Jacksonville, FL* **July 2024 – November 2024**

- Reduce payroll process time by 75% for 23 virtual assistants using Excel for #1 Turo cohost at *RentNiceCars.com*

Electrical Engineering Intern | *The Haskell Company - Jacksonville, FL* **June 2022 – July 2022**

- Communicate professionally among engineers and learn basics of electrical design and project documentation

PROJECTS

Soccer Rover **June 2025 – September 2025**

Developed for High School Robotics Camp at UC Irvine Samueli School of Engineering

I took full responsibility of creating and teaching an autonomous soccer rover project. I built the project with Raspberry Pi 3, 3D prints, a laser cut chassis, motors, a power system, and a camera module. Open CV and Python were used for directional logic. Within one week, 28 students had their very own soccer rover. (See Github for more)

Optical Neural Network Research **May 2025 – February 2026**

Shcherbakov Nanophotonics Lab at UC Irvine

Researched the translation of Convolutional Neural Networks from slower digital computation to a speed of light Optical Neural Network using Spatial Light Modulation in a 4f System along with nonlinear optical responses.

Embedded Systems and Robotics Projects at UC Irvine **September 2023 – December 2025**

- **Remote-Controlled Rover** with NRF24L01 radio, Arduino Nano, L293D Driver, and a custom PCB design
- **Wrist-wearable Fitness Tracker** with accelerometer, barometer, WEMOS dev board, and Blynk app connection
- **ArduPilot-based quadcopter** with GPS, flight controller, electronic speed controllers, and wireless telemetry capable of autonomous flight missions using Mission Planner.

Personal Activities: Piano, Volleyball, Tennis, Biking, Volunteering at Church, Raspberry Pi Tinkering, Fighting Games

github.com/ajcalinisan · linkedin.com/in/aden-calinisan